

Contact Information



Team

Sungmin Kim | Founder & CEO

- Physics, KMU
- Patent attorney / KCVA

Dongwoo An | Business

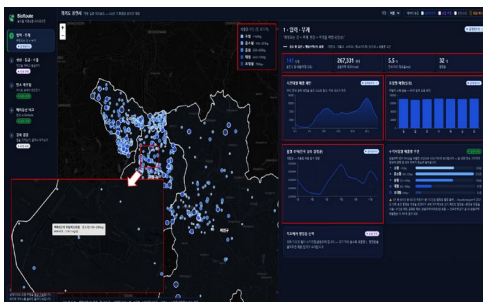
- Business Administration, SNU
- Finance and growth strategy

Hyunsung Shim | Engineering

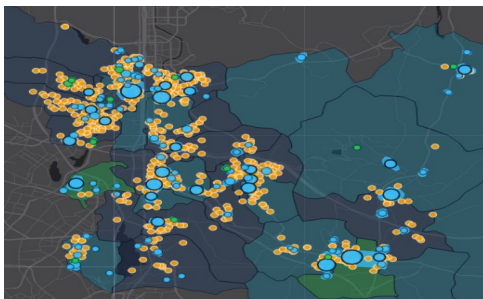
- Electrical Engineering, KAIST
- Estimation, validation and data integration

Byung-il Jang | Advisor

- Professor, KAIST GSGS
- Waste, energy and policy advisory



Public RFID discharge records



Estimated composition grades



Illustrative collection route

Problem

- Korea's Biogas Act makes 243 municipalities convert 50% of organic waste to biogas, with penalties for shortfalls.
- Only 6.6% of organic waste becomes energy today; the same ton emits 25× more methane depending on the route.
- Nobody measures what is in the waste, so nobody decides its route.

Solution

- BioRoute-Food Waste estimates waste composition from public data alone: discharge records, housing, households, nearby businesses.
- It grades each collection point by methane potential and routes high-yield waste unmixed to anaerobic digestion.
- Same waste, more energy, lower penalties. No new sensors or plants.

Key Features

- Every coefficient is published and traceable, so a city official can audit exactly why a district received grade B.
- Compliance shown in January, not in a December notice : target vs. expected production, penalty exposure.
- One engine, three market roles : built in Korea, tested in Singapore, then Indonesia and Thailand.

Business Model

- B2G success fee: 20–30% of avoided penalty, plus an annual subscription for the routing and reporting dashboard.
- Gain-share with facility operators on co-digestion yield.
- Korea: 47 target cities, about \$46–62M annual penalty exposure (estimate). Similar regulatory pull in US, EU, Singapore.

Business Snapshot

- Bootstrapped to date; selected for Ministry of SMEs commercialization support (2026)
- KAIST Climate-Tech Audition, Excellence Award (Sep 2026);
- Yongin concept demonstration
- Validation-first MVP in development, benchmarked on five field studies
- Pre-revenue; next 12 months: Yongin field validation, a Singapore test, then Indonesia or Thailand

Target Audience

- Municipalities and public facility operators bound by Korea's Biogas Act
- Singapore: large malls under NEA waste reporting, and the partners who collect or treat their waste
- Indonesia and Thailand: cities assessing feedstock for new digesters

Seeking Opportunities

- Pilot partners with a facility or a waste stream, Singapore first
- Validation partners in Singapore and Southeast Asia
- Pre-seed investors and accelerator partners